Presenting Research Plans and Proposals

LS100 — Module 00A · Research Plans & Proposals

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This guide focuses on the first presentation for LS100 students, not the written proposal, which is covered separately. It is intended for the 15-minute group talk you give at the very start of the semester. It assumes you have already worked through *Research Methodologies: Converting the Problem to a Research Framework* (Guide 00) in this module — this talk presents that framework rather than builds it.

In this talk, you do not need to show results as you have not run the study yet. It is rather intended to present your idea, the question or problem you want to solve, and the research plan in a coherent way, and getting feedback before you begin the research. Build the talk to invite questions, not avoid them. Keep it to a maximum of **four slides** — below is what each slide may cover, and how to present well.

1. Your four slides

Slide	Content
1	The overarching question — and why it matters
2	Your specific aims and hypotheses / target
3	Methodology and data — how each dataset answers an aim / serves the target
4	Possible outcomes, and back to the bigger question

This structure mirrors how NIH grant applications begin: a one-page "Specific Aims" section that states the problem's significance, lays out the aims and hypotheses, and closes by pointing to the expected impact ((National Institute of Allergy and Infectious Diseases (NIAID), n.d.)).

2. 1. Defining and framing the problem

Slide 1

- Open with the overarching issue.
- Explain why the problem matters to you and why it may matter to a broader audience — especially if the full research would require substantial funding.

- From the broad topic/ issue, narrow down to one specific, testable question. ("What drives X" is a topic; "Does A predict B in Y" is a question.)
- **Avoid asking tautological questions** - questions whose answer is guaranteed by the definitions involved, and no study could ever come out with different results. For example, "Does arm kinematics differ across different types of baseball or golf swings?" — logically, it must. No amount of study or data will find anything otherwise. Therefore, the question is not interesting. Instead, ask questions that investigate empirical unknowns, where the outcomes are not predetermined, and the hypotheses can be actively tested and falsified.
- Keep the question or problem statement to one sentence that your audience could repeat back to you. If they cannot, it is not yet ready for slide 1.
- Ideally, frame the research question by pointing out the gap: what is already known and what remains unresolved. For this presentation, that may depend on how much time you want to allocate to this slide.

For the talk: Open with a short story — what's known or agreed on (the "and"), the tension or gap (the "but"), and what you propose to do about it (the "therefore"). One well-crafted sentence like this can carry the whole slide ((Olson, 2015)).

= 2. Breaking down the problem: specific aims and hypotheses

Slide 2

- To solve the problem, what related questions or steps must be addressed, either in sequence or in parallel? These might include one specific research question, one testable hypothesis, or a deliverable outcome - developing a tool, or a statistical or machine learning model.
- State each as a specific aim: what you will do, and what you expect to find. Each aim (question or step) should lead to a clear hypothesis or outcome. Think of hypothesis as a specific, testable, falsifiable expectation — "if [X], then [Y], because [mechanism/ logic]" is a good template. Outcome could be a tool, data pipeline, a statistical or machine learning model.
- Check whether each aim is feasible within the time and resources you actually have.
- If a later aim depends entirely on an earlier one succeeding, say so and name it as a risk.
- Write down the questions, hypotheses, steps, and possible outcomes in a logical sequence. This helps you to have a mental map of the logical flow.
- This process is iterative, so you may need to revisit it and add, remove, or reorder items — this is a core part of research.

For the talk: Keep this to two aims. Limiting the presentation to two aims makes the project feel focused rather than unfocused, even when the underlying

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research is ambitious. This is what NIH suggests to the grant writers as well ((National Institute of Allergy and Infectious Diseases (NIAID), n.d.)).

= 3. Focusing on one achievable goal: Methodology and data structure

Slide 3

- Next, focus on the first aim. Is it achievable within your available time and resources? If yes, nice. If not, limit it within your budget.
- Name every dataset you will need, specify its type, and identify the relevant variables (for example, independent vs. dependent variables, and whether they are discrete or continuous). Then draw a direct line from each dataset to the aim it answers.
- Explain where the data will come from, what equipment or access you will require to get the data, and how long it will take to collect the data. As a rule of thumb, **consider budgeting about a third of your available time for data collection**.
- If you cannot measure a construct directly, name the proxy you will use (for example, a click count on a website standing in for "engagement"), and explain in one sentence why it is a reasonable stand-in and one way it could fail (a threat to construct validity (Cronbach & Meehl, 1955)).
- Briefly note any transformation the raw data will need before analysis, and the statistical or computational method you will use.

For the talk: You may show this as a diagram or infographics - for example, a flow showing how each specific aim connects to the relevant dataset and method. A figure someone else could present without you standing there is doing its job; if it takes more than a few seconds to read, simplify it ((Midway, 2020)).

= 4. Possible outcomes and the bigger picture

Slide 4

- State the outcome that would tell you the analysis worked — for example, a hypothesis rejected, or a pipeline produced, or a model trained.
- Connect this specific aim back to the larger theoretical, practical, or societal question you opened with, and name the next question that follows from it.
- Say what you will do with the feedback from this talk, keeping in mind that this presentation is a proposal, not a conclusion.

For the talk: Close by answering the "who cares" question you opened with. Compressing a research question and its stakes for a general audience in a few minutes is a real, transferable skill that I would like you to gain from this exercise.

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= Delivering it well

- **Rehearse out loud against a timer**. If you are presenting in group (which is the case for LS100 students), calculate how much time each of you will have. However, each of you should have at least 3 minutes - that is 180 seconds. So, each second counts.
- When someone asks a question, repeat or rephrase it before you answer — it confirms you understood it, and makes sure the room heard it too.
- Build the talk to invite questions, not close them off: ask yourself what you want your audience to ask, not just what you need to tell them ((Heath & Heath, 2007)). The more engaged your listeners are while you talk, the more they understand and remember afterward ((Stephens et al., 2010)).

3. Before you present: a quick self-check

- Slide 1: Can someone outside the behavioral sciences follow your one-sentence question?
- Slide 2: Are your aims specific and falsifiable, not just topics?
- Slide 3: Does every dataset trace to an aim, and does the timeline add up?
- Slide 4: Have you named the outcome that tells you it worked, and closed the loop on "why it matters"?
- Have you timed a full run-through, reading out loud, at four slides? If it exceeds 3 minutes, revise it.

References

Cronbach, L. J., & Meehl, P. E. (1955). Construct validity in psychological tests. *Psychological Bulletin*, *52*(4), 281–302. https://doi.org/10.1037/h0040957

Heath, C., & Heath, D. (2007). *Made to Stick: Why Some Ideas Survive and Others Die*. Random House.

Midway, S. R. (2020). Principles of effective data visualization. *Patterns*, *1*(9), 100141. https://doi.org/10.1016/j.patter.2020.100141

National Institute of Allergy and Infectious Diseases (NIAID). (n.d.). *Write Your Research Plan*. Retrieved https://www.niaid.nih.gov/grants-contracts/write-research-plan

Olson, R. (2015). *Houston, We Have a Narrative: Why Science Needs Story*. University of Chicago Press.

Stephens, G. J., Silbert, L. J., & Hasson, U. (2010). Speaker–listener neural coupling underlies successful communication. *Proceedings of the National Academy of Sciences*, *107*(32), 14425–14430. https://doi.org/10.1073/pnas.1008662107